Summary in Graph

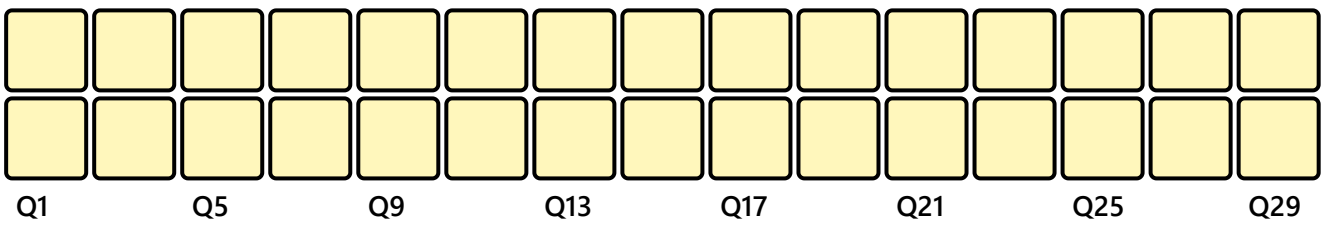
Exam Summary (GATE Overflow CS Test Series 2026 | Computer Organisation and Architecture | Test 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (7) out of 71 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 CO & Architecture

The correct sequence of steps involved in the execution of the instruction LOAD R1, X(R2) are

a. Fetch instruction and increment PC.
b. Compute sum [R1] + [R2]
c. Decode instruction to determine operation.
d. Keep quiet (No memory access)

e. Add immediate value to the content of R2.
f. Use sum X+[R2] as an effective address of source operand and read content.
g. Load the data received from the memory into destination register R1
h. Load register's result into destination register R1

A. a, c, e, f, g
B. a, b, c, d, e
C. a, b, e, f, h
D. a, e, f, g, h

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

Which one of the following is correct?

A. If Interrupt enable bit is 1 in CPU, then processor is interrupt safe.
B. Daisy chain is less efficient than software polling which provides in effect a hardware poll and for interrupts all I/O modules share a common interrupt request line.
C. In software polling when the processor detects an interrupt, it branches to an ISR whose job is to poll each module to determine which module caused the interrupt
D. All options are correct.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Numerical Type | Award: 1 | Penalty: 0 | CO & Architecture

The Fetch, Decode, Execute, Memory and Write Back stages of a pipelined processor have the latencies 200ps, 140ps, 160ps, 190ps and 100ps respectively. Assume that when pipelining, each pipeline stage costs 10ps extra for the registers between pipeline stages. If you could split one of the pipeline stages into 2 equal halves, what is the new latency (in ps) for an instruction? (rounded to one decimal point)

Your Answer: | Correct Answer: 1200 : 1201 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | CO & Architecture

Which one of the following is/are INCORRECT? (Mark all the appropriate choices)

A. Compulsory misses can be reduced by increasing the total cache size.
B. Capacity misses can be reduced by increasing the block size.
C. Conflict misses may be increased by increasing the value of associativity.
D. Compulsory misses can be reduced by increasing the cache block size.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

A computer system has a $8\ K$ word cache organized in block-set-associative manner with 8 blocks per set and 32 words per block. The number of bits in the SET and WORD fields of the main memory address format is:

A. $6, 4$

B. $4, 5$
C. $5, 5$
D. $8, 5$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

The main memory of a computer has N blocks while the cache has N/m blocks. If the cache uses set associative mapping scheme with 4 blocks per set, then block k of the main memory maps to the set:

A. $(k \mod (N/m))$ of the cache
B. $(k \mod (N/4m))$ of the cache
C. $(4k \mod 4m)$ of the cache
D. $(k \mod 4m)$ of the cache

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

Which of the following is/are correct?

a. In RISC, control circuit is simpler than CISC kind of architectures
b. In vertical microprogramming, control signals cannot be directly generated after fetching of microinstructions
c. Horizontal microprogrammed control unit requires longer control word compared to vertical microprogramming
d. Horizontal microprogrammed control unit is more flexible than vertical microprogrammed.

A. (a)(b)(c)(d)
B. (a)(b)(c)
C. (a)(b)
D. (b)

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Numerical Type | Award: 1 | Penalty: 0 | CO & Architecture

Consider the following Assembly code fragment

```
ldr r0, adr_var1
    @ load the memory address of var1 via
    label adr_var1 into r0
ldr r1, adr_var2
    @ load the memory address of var2 via
    label adr_var2 into r1
ldr r2, [r0]
    @ load the value at memory address
    found in r0 to register r2
str r2, [r1]
    @ store the value found in r2
    to the memory address found in r1
```

The total number of memory accesses (excluding register accesses) during the execution stage when the above sequence of instructions is executed is _____

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

Two computers use register windows with the following characteristics. Determine which computer is using more number of the total number of registers and find their absolute difference.

	Computer A	Computer B
Global registers	10	9
Local registers	9	8
Common registers	6	7
Number of windows	8	4

A. $B, 130$
B. $A, 69$
C. $A, 61$
D. $B, 61$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

Consider the following one address instruction for a hypothetical processor implementing a function:

```
    MOVE x
    MUL x
    STORE z
    MOVE x
5.  MUL A
    ADD z
    ADD B
    STORE y
```

Which of the following values represents the value of the function if $A = 7, B = 6$ and input value $x = 10_{16}$ (Hexadecimal)?

A. 347_{10}
B. 176_{10}
C. 167_{10}
D. 374_{10}

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

A computer using expanding opcode technique has $32 - bit$ instructions and $13 - bit$ addresses. If there are 60 two-address instructions, and 128 zero-address instructions, approximately how many one-address instructions can it support?

A. $32\ K$
B. $20\ K$
C. $64\ K$

D. $16\ K$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

The content of the top of memory stack is 5230. The content of the stack pointer SP is 2560. A two word call subroutine instruction is located in memory at address 1020 followed by the address field of 6720 at location 1021. If PC value before the call instruction is fetched from memory is A, PC value immediately after the call instruction is executed is B and PC value immediately after the return from subroutine is C then value of tuple $\langle A, B, C\rangle$ is

A. $\langle 1021, 5230, 1021\rangle$
B. $\langle 1021, 6720, 1022\rangle$
C. $\langle 1020, 5230, 1022\rangle$
D. None of the above

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

A processor that has carry, overflow and sign flag bits as part of its program status word (PSW) performs addition of the following two $2's$ complement numbers 01101100 and 11000011. After the execution of this addition operation, the status of the carry, overflow and sign flags, respectively will be:

A. $0, 1, 1$
B. $1, 0, 1$
C. $0, 1, 0$
D. $1, 0, 0$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Consider a CPU where all the instructions require 8 clock cycles to complete execution. There are 150 instructions in the instruction set. It is found that 132 control signals are needed to be generated by the control unit. While designing the horizontal microprogrammed control unit, single address field format is used for branch control logic. What is the minimum size of the control word and control address register (in bits)?

A. $134, 10$
B. $143, 10$
C. $143, 11$
D. $134, 11$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 01sec | Discuss

Q #15 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

A non-pipeline processor X has a clock frequency of $2.5\ GHz$ and an average CPI of 3. Processor Y an improved version of X, is designed with 5 stage linear instruction pipeline. However, due to latch delay and clock skew the clock rate of Y is only $2\ GHz$. If a program consists of one million instructions are executed on both the processors the speedup of processor Y as compared to X is _____ (rounded to 1 decimal points)

Your Answer: | Correct Answer: 2.4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Which option is the correct order for transfer of a block of data?

a. The peripheral device (such as disk controller) will request the service of DMA by pulling DREQ high
b. CPU will finish the present bus cycle (not necessary the present instruction) and respond to the DMA by putting high on its HLDA.
c. DMA will put high on its HRQ, signalling the CPU through its HOLD pin that it needs to use the buses.
d. DMA will activate DACK which tells peripheral device that it will start to transfer the data.
e. DMA starts transferring from memory to peripheral by putting the address of first byte of block on the address bus and activating MEMR.
f. DMA decrements the counter and increment the address points and repeats until count zero.
g. When finished deactivate HRQ now it can regain control over its buses.

A. $(a)(c)(b)(d)(e)(f)(g)$
B. $(a)(b)(d)(e)(f)(c)(g)$
C. $(a)(b)(c)(d)(e)(f)(g)$
D. $(c)(b)(a)(d)(e)(f)(g)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

An instruction pipeline consists of 5 stages - Fetch (F), Decode (D), Execute (E), Memory (M) and Write (W). The four instruction in an instruction sequence need these stages for different number of clock cycles as shown below:

Instruction	F	D	E	M	W
1	1	2	1	2	1
2	2	2	2	1	1
3	1	1	3	2	2
4	1	3	2	1	1

The number of clock cycles needed to perform these four instructions is _____

Your Answer: | Correct Answer: 15 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Consider a 5-stage pipeline - IF (Instruction Fetch), ID (Instruction Decode and register read), EX (Execute), MEM (Memory) and WB (Write Back). All register reads take place in the second phase of a clock cycle and all register writes occur in the first phase. Consider the execution of the following instruction sequence:

- $I_1 : R_1 \leftarrow R_2 + R_3$
- $I_2 : R_3 \leftarrow R_1 - R_2$
- $I_3 : M[R_1 + 1000] \leftarrow R_1$

- $I_4 : R_2 \leftarrow R_3 * R_1$

If the number of RAW (Read after write) hazards is denoted by A, WAR (Write after read) hazards by B and WAW (Write after write) hazards by C, then $A + B + C =$____

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Suppose the functions A and B can be computed in 6 and 4 nanoseconds by functional units U_A and U_B, respectively. Given two instances of U_A and three instances of U_B, it is required to implement the computation $A(B(X_i))$ for $1 \leq i \leq 100$. Ignoring all other delays, the minimum time required to complete this computation (in nanoseconds) is _____

Your Answer: | Correct Answer: 304 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Consider an instruction pipeline with five stages without any branch prediction: Fetch Instruction(FI), Decode Instruction(DI), Fetch Operand(FO), Execute instruction(EI) and Write Operand(WO). The stage delays for FI, DI, FO, EI and WO are $4ns, 5ns, 12ns, 7ns$ and $6ns$ respectively. There are intermediate storage buffers after each stage and the delay of each buffer is 1ns. A program consisting of 12 instructions $I_1, I_2, \ldots I_{12}$ is executed in this pipelined processor. Instruction I_4 is the only branch instruction and its branch target is I_{10}. If the branch is taken during the execution of this program, the time(in ns) needed to complete the program is _____

Your Answer: | Correct Answer: 182 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

In a C program, an array is declared as $\text{double arr}[8192]$. The starting address of the array is 0x00000000. This program is run on a computer that has a 4-way set associative cache of size 32 Kbytes, with block size of 64 Bytes. If the program accesses the elements of the array one by one in order from first element to last and then one more time in reverse order from the last element to the first element, total number of cache misses will be _____

(Assume that the data cache is initially empty and that no other data or instruction accesses are to be considered.)

A. 2048
B. 1536
C. 256
D. 1024

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

A Disk drive with 8 surfaces, 256 tracks/surface, 256 sectors/track, 512 bytes/sector and rotation speed of 3600 rpm is operated in cycle stealing mode whereby whenever one 2 Byte word is ready, it is sent to memory. Similarly for writing, the disk interface reads a 2 Byte word from memory in each DMA cycle. The memory cycle time is 20 nanoseconds. Find the maximum percentage of time that a CPU gets blocked during

DMA operations (rounded to 1 decimal point).

Your Answer: | Correct Answer: 7.6 : 7.9 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

A Hard disk is connected to a 2 GHz processor through a DMA controller. Assume that the initial setup time for a DMA transfer takes 1000 clock cycles for the processor and also assume that the handling of the interrupt on DMA completion requires 500 clock cycles for the processor. The hard disk has a transfer rate of 2000 MBps and average block size transferred is 4 KB. What amount of the processor time is consumed by the disk, assuming that the data are transferred only during the idle cycles of the CPU?

A. 2.060 milliseconds
B. 30 microseconds
C. 1.020 microseconds
D. None of the above

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

On a non-pipelined sequential processor, a program segment, which is the part of the interrupt service routine, is given to transfer 400 bytes from an I/O device to memory.

```
        Initialize the address register
        Initialize the count to 400
LOOP:   Load a byte from device
        Store in memory at address given by address register
        Increment the address register
        Decrement the count
        If count!=0 go to LOOP
```

Assume that each statement in this program is equivalent to a machine instruction which takes one clock cycle to execute if it is a non-load/store instruction. The load/store instruction take 3 clock cycles to execute. The designer of the system also has an alternate approach of using the DMA controller to implement the same transfer which requires 30 clock cycles of initialization and other overheads. Each DMA transfer cycle takes two clock cycles to transfer one byte of data from the device to the memory. What is the approximate speedup when the DMA controller based design is used in a place of interrupt driven program based input-output? (round off to 2 decimal places)

Your Answer: | Correct Answer: 4.33 : 4.34 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Consider a direct mapped cache of size 16 KB and block size 64 bytes and using LRU replacement. Initially the cache is empty. The following sequence of access to memory blocks :

$$0, 64, 0, 288, 320, 16, 34, 257, 274, 19, 12, 32, 1$$

is repeated 100 times. If the number of conflict misses and compulsory misses are A and B respectively, $2A + B =$ ____

Your Answer: | Correct Answer: 1200 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Assume a main memory access time of $64\ ns$ and a memory system capable of a sustained transfer rate of 16 GBps. If the block size is 64 bytes, what is the maximum number of outstanding misses we need to support assuming that we can maintain the peak bandwidth given the request stream and that accesses never conflict. For simplicity, ignore the time between misses.

Your Answer: | Correct Answer: 16 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

For the C code below, determine the number of data cache misses assuming an 8 KB direct-mapped data cache with 64-byte blocks, and a write-back cache that does write allocate. The elements of a and b are 4 bytes. Let's also assume they are not in the cache at the start of the program and starting addresses of a and b are at cache line boundaries.

```
int a[4][96], b[97][4];
....
for (i = 0; i < 4; i++)
    for (j = 0; j < 96; j++)
        a[i][j] = b[j][0] * b[j+1][0];
```

Your Answer: | Correct Answer: 49 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Consider a cache of size 256 KB, 4-way set associative, with a block size of 64 bytes. If the byte addressable main memory size is $4\ GB$, what is the tag memory size for the cache in bytes?

Your Answer: | Correct Answer: 8192 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

The Fetch, Decode, Execute, Memory and Write Back stages of a processor have the latencies $300ps, 300ps, 350ps, 500ps$ and $110ps$ respectively. What is the ratio of the throughputs of a pipelined implementation with an extra overhead of $20ps$ for the register between the pipeline stages and a non-pipelined implementation (correct to one decimal place)?

Your Answer: | Correct Answer: 3.0 : 3.1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Consider a CPU with an average CPI of 1.4 when all memory accesses hit on the cache. Assume an instruction mix

ALU	45%
LOAD	20%
STORE	20%
BRANCH	15%

If the cache miss rate is 1.6% and miss penalty is 60 cycles, calculate the effective CPI for a unified $L1$ cache using write back and write allocate policy with the probability of a cache block being dirty being 0.15 (round off to 2 decimal places).

Your Answer: | Correct Answer: 2.94 : 2.95 | Not Attempted | Time taken: 00min 00sec | Discuss

Copyright & Stuff